AS · Unit 2 · WCH12 Energetics, Group Chemistry, Halogenoalkanes and Alcohols

Energetics

Every chemical reaction takes in or gives out energy, because bonds have to be broken and new ones made. This topic shows you how to define, measure and calculate these enthalpy changes: with a thermometer and a polystyrene cup, with Hess cycles that find values nobody can measure directly, and with bond enthalpies.

Enthalpy changes

In energetics we split the world into two parts. The **system** is the chemicals that are reacting. The **surroundings** are everything else: the water they are dissolved in, the beaker, the thermometer, the air. When a reaction happens, energy moves between the two, and the total amount of energy stays the same.

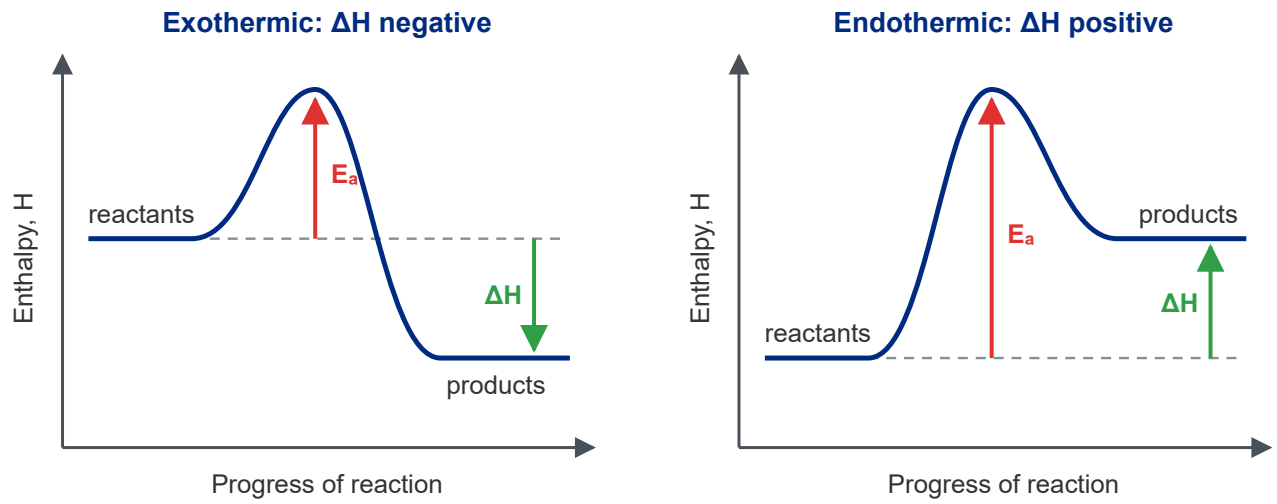
The heat content of a system is called its **enthalpy**, H. Nobody can measure the enthalpy of a substance on its own, but we can measure how much it changes. The **enthalpy change**, ΔH, is the heat energy transferred between the system and the surroundings when a reaction takes place at **constant pressure**. Most reactions in a lab happen in open containers, at the pressure of the atmosphere, so this is the quantity we actually measure.

- **Exothermic:** the system gives heat energy out to the surroundings, so the surroundings warm up. The products have less enthalpy than the reactants, so ΔH is **negative**. Combustion, neutralisation and respiration are all exothermic.
- **Endothermic:** the system takes heat energy in from the surroundings, so the surroundings cool down (or the reaction needs continuous heating). The products have more enthalpy than the reactants, so ΔH is **positive**. Thermal decomposition, such as $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$, and photosynthesis are endothermic.

Why do reactions transfer energy at all? Breaking a bond always **takes in** energy, because you have to pull the atoms apart against the attraction between the bonding electrons and the two nuclei. Making a bond always **gives out** energy. If the bonds made in the products are stronger overall than the bonds broken in the reactants, more energy comes out than goes in and the reaction is exothermic. If the bonds broken are stronger overall, the reaction is endothermic.

Enthalpy level diagrams and reaction profiles

An enthalpy level diagram puts the reactants and products at heights that show their enthalpy. The ΔH arrow always starts at the reactants and points to the products, so it points **down** for an exothermic reaction and **up** for an endothermic one. A reaction profile adds the path between them. Even an exothermic reaction has to climb a hump first: some bonds must start to break before new ones can form. The height of that hump above the reactants is the **activation energy**, E_a, the minimum energy that colliding particles need for a reaction to happen.



Reaction profiles. The activation energy is measured from the reactants up to the top of the hump. ΔH is the difference between the reactants and the products, whatever happens in between.

EXAM TIP

When you draw an enthalpy level diagram, label the vertical axis "enthalpy", write the formulas of the reactants and products on their levels, and draw the ΔH arrow **from reactants to products** with a single arrowhead. An exam answer that has the arrow pointing the wrong way loses the mark even if the levels are right.

Standard conditions

The size of an enthalpy change depends on the temperature, the pressure, the concentrations and the physical states of the substances. Water vapour condensing to liquid water gives out extra energy, for example, so burning methane to make $H_2O(l)$ gives a bigger ΔH than burning it to make $H_2O(g)$. To compare values fairly, chemists quote them under **standard conditions** and mark them with the symbol $\ominus$, as in $\Delta H^{\ominus}$.

- **Pressure:** 100 kPa.
- **Temperature:** a stated temperature, normally 298 K (25 °C).
- **Solutions:** concentration 1.00 mol dm^{-3}.
- **Standard states:** every substance is in the physical state it normally has at 100 kPa and 298 K. Water is $H_2O(l)$, oxygen is $O_2(g)$, bromine is $Br_2(l)$, iodine is $I_2(s)$, carbon is C(s) in the form of graphite.

ΔH values are quoted in kJ mol^{-1}. The "per mole" refers to the equation as written: if you double every number in the equation, you double ΔH.

Standard enthalpy change definitions

You need to know five definitions word for word in meaning. Each one fixes three things: **what reacts**, **how much** (usually exactly one mole of something), and the **conditions and states**.

DEFINITION

Standard enthalpy change of reaction, $\Delta_r H^{\ominus}$

The enthalpy change when the amounts of reactants shown in the equation react together under standard conditions, with all substances in their standard states.

DEFINITION

Standard enthalpy change of formation, $\Delta_f H^{\ominus}$

The enthalpy change when **one mole** of a compound is formed from its **elements** in their standard states, under standard conditions.

The equation must make exactly one mole of the compound, which often means fractions in front of the elements:

$$2C(s) + 3H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow C_2H_5OH(l) \quad \Delta_f H^{\ominus} = -277 \text{ kJ mol}^{-1}$$

$$Na(s) + \tfrac{1}{2}Cl_2(g) \rightarrow NaCl(s) \quad \Delta_f H^{\ominus} = -411 \text{ kJ mol}^{-1}$$

The standard enthalpy of formation of an **element in its standard state is zero**. Forming $O_2(g)$ from $O_2(g)$ is no change at all. This is not true of other forms of the element: $O_3(g)$ or $Br_2(g)$ have non-zero values.

DEFINITION

Standard enthalpy change of combustion, $\Delta_c H^{\ominus}$

The enthalpy change when **one mole** of a substance is burned **completely in excess oxygen** under standard conditions, with all reactants and products in their standard states.

$$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l) \quad \Delta_c H^{\ominus} = -1367 \text{ kJ mol}^{-1}$$

Combustion is always exothermic. Complete combustion of a compound of carbon and hydrogen gives only CO_2 and H_2O. With too little oxygen you get incomplete combustion, making carbon monoxide and soot (carbon). Less energy is released, because the carbon has not been fully oxidised. Notice also that the combustion of hydrogen, $H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(l)$, is the same reaction as the formation of water, so $\Delta_c H^{\ominus}(H_2) = \Delta_f H^{\ominus}(H_2O(l)) = -286$ kJ mol^{-1}. In the same way, $\Delta_c H^{\ominus}$(C, graphite) $= \Delta_f H^{\ominus}(CO_2) = -394$ kJ mol^{-1}.

DEFINITION

Standard enthalpy change of neutralisation, $\Delta_{neut}H^{\ominus}$

The enthalpy change when an acid and an alkali react together in solution to form **one mole of water**, under standard conditions.

For any strong acid with any strong alkali the value is about −57 kJ mol^{-1}. The reason is that strong acids and alkalis are fully ionised, so the only reaction that actually happens is the same every time:

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$$

The other ions (Na^+, Cl^-, K^+, NO_3^- and so on) are spectators. A weak acid such as ethanoic acid gives a slightly less exothermic value, because some energy is used up in ionising the acid as it reacts. The definition is per mole of **water**: $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$ makes two moles of water, so the enthalpy change for that equation is about twice $\Delta_{neut}H^{\ominus}$.

DEFINITION

Standard enthalpy change of atomisation, $\Delta_{at}H^{\ominus}$

The enthalpy change when **one mole of gaseous atoms** is formed from an element in its standard state, under standard conditions.

$$Na(s) \rightarrow Na(g) \quad \Delta_{at}H^{\ominus} = +107 \text{ kJ mol}^{-1}$$
$$\tfrac{1}{2}Cl_2(g) \rightarrow Cl(g) \quad \Delta_{at}H^{\ominus} = +122 \text{ kJ mol}^{-1}$$

Atomisation is always endothermic, because bonds (metallic bonds in sodium, the covalent Cl–Cl bond in chlorine) have to be broken. For chlorine, the definition is per mole of Cl *atoms*, so it is half the energy needed to break one mole of Cl–Cl bonds.

Enthalpy change	Fixed amount	Sign
reaction, $\Delta_rH^{\ominus}$	the molar amounts in the equation as written	either
formation, $\Delta_fH^{\ominus}$	1 mol of compound made from its elements	usually negative
combustion, $\Delta_cH^{\ominus}$	1 mol of substance burned completely	always negative
neutralisation, $\Delta_{neut}H^{\ominus}$	1 mol of water formed	always negative
atomisation, $\Delta_{at}H^{\ominus}$	1 mol of gaseous atoms formed	always positive

COMMON MISTAKE

Two common slips in definitions: forgetting the state symbols, and getting the "one mole" wrong. $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$ is *not* the formation equation of water, because it makes two moles. And $H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow$

$H_2O(g)$ is not either, because water's standard state is liquid.

Measuring enthalpy changes: calorimetry

You can't measure the heat a reaction gives out directly. What you can measure is its effect: the temperature change of a known mass of water or solution that absorbs the heat. The energy needed to warm a substance depends on its mass, on how far its temperature changes, and on how much energy that substance needs per gram per degree, its **specific heat capacity**, c.

$$Q = mc\,\Delta T$$

- **Q:** heat energy transferred, in J.
- **m:** the mass of the water or solution whose temperature changes, in g. Not the mass of the reactants, and not the mass of any solid added.
- **c:** specific heat capacity. For water and dilute solutions, use 4.18 J g^{-1} K^{-1}.
- **ΔT:** temperature change. A change of 1 K is the same size as a change of 1 °C, so either scale works for ΔT.

Q is the energy for the amounts you actually used. To turn it into ΔH you divide by the number of moles of the substance that sets the amount of reaction, then convert J into kJ and add the sign: a temperature **rise** means the reaction gave out heat, so ΔH is **negative**.

$$\Delta H = \frac{-Q}{n}$$

Reactions in solution

For neutralisation, displacement, dissolving and similar reactions, the solution itself absorbs the heat. The reaction is done in an expanded polystyrene cup, which is a good insulator and has a very small heat capacity, so almost all the energy stays in the solution.

- Stand the polystyrene cup in a beaker for support and extra insulation, and use a lid with a hole for the thermometer.
- Measure the solution into the cup with a pipette or burette. Weigh any solid reactant accurately, preferably by weighing the container before and after adding the solid.
- Record the temperature every minute for 3 minutes to get a steady starting value. If two solutions are mixed, measure both and use the average.
- At the fourth minute add the second reactant, stir, and keep recording every half-minute or minute for several more minutes. Stirring makes sure all of the solution is at the same temperature.
- Plot temperature against time and **extrapolate** the cooling line back to the moment of mixing to find ΔT.

Why extrapolate? While the reaction is still giving out heat, the warm solution is already losing some to the room. The highest temperature you read is therefore lower than it would have been if all the heat had been released instantly. The steady cooling afterwards shows how fast heat is being lost; following that line back to the time of

mixing estimates the temperature the solution would have reached with no losses. The slower the reaction, the more this matters.

Worked example: a displacement reaction

1. 50.0 cm^3 of 0.500 mol dm^{-3} copper(II) sulfate solution is placed in a polystyrene cup and an excess of zinc powder is added. The extrapolated temperature rise is 24.5 °C. Calculate the enthalpy change for $Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$.

2. Heat absorbed by the solution. The mass is that of 50.0 cm^3 of solution, taken as 50.0 g:

$$Q = mc\Delta T = 50.0 \times 4.18 \times 24.5 = 5120.5 \text{ J}$$

The zinc is a solid that does not get counted in m. We also assume the solution has the density and heat capacity of water.

3. Moles of the reactant that is not in excess, copper(II) sulfate:

$$n = cV = 0.500 \times \frac{50.0}{1000} = 0.0250 \text{ mol}$$

4. Divide, convert to kJ, and give it a negative sign because the temperature rose:

$$\Delta_r H = \frac{-5120.5}{0.0250} = -204\,820 \text{ J mol}^{-1} = -205 \text{ kJ mol}^{-1}$$

The data book value is −217 kJ mol^{-1}. The experiment gives a smaller magnitude because some heat still escaped and some warmed the cup and thermometer.

Worked example: neutralisation

1. 50.0 cm^3 of 1.00 mol dm^{-3} hydrochloric acid and 50.0 cm^3 of 1.00 mol dm^{-3} sodium hydroxide solution, both at 19.8 °C, are mixed. The extrapolated maximum temperature is 26.4 °C. Calculate the enthalpy change of neutralisation.

2. Both solutions end up warmer, so the mass is the total volume, 100.0 cm^3 = 100.0 g. ΔT = 26.4 − 19.8 = 6.6 K.

$$Q = 100.0 \times 4.18 \times 6.6 = 2758.8 \text{ J}$$

3. The equation is HCl + NaOH → NaCl + H_2O, so the moles of water formed equal the moles of acid (and of alkali):

$$n(\mathrm{H_2O}) = 1.00 \times \frac{50.0}{1000} = 0.0500 \text{ mol}$$

4. $$\Delta_{\text{neut}}H = \frac{-2758.8}{0.0500} = -55\,176 \text{ J mol}^{-1} = -55.2 \text{ kJ mol}^{-1}$$

Always finish with a sign, units and a sensible number of significant figures (normally 3, or fewer if the data have fewer).

COMMON MISTAKE

The most common errors in these calculations: using only one solution's volume for *m* when two solutions are mixed, adding the mass of a solid to *m*, forgetting to divide by 1000 to get kJ, and leaving off the minus sign for an exothermic reaction.

Sources of error and assumptions

- **Heat loss** (or gain) to the surroundings, the biggest error. It makes the measured ΔH less exothermic (or less endothermic) than the true value.
- **Heat capacity of the apparatus** is ignored: the cup and thermometer absorb a little energy too.
- **The solution is treated as water**: density 1.00 g cm^{-3} and c = 4.18 J g^{-1} K^{-1}.
- **The reaction may be incomplete or slow**, or a solid may not fully dissolve.
- **Uncertainty in ΔT:** a thermometer reading has an uncertainty of half a division, and ΔT uses two readings. A larger temperature change (more concentrated solutions, or more solid) makes the *percentage* uncertainty in ΔT smaller.

Enthalpy change of combustion

To measure $\Delta_c H$ of a liquid fuel such as an alcohol, burn it in a spirit burner under a copper can containing a known mass of water. Weigh the burner (with its cap on, to stop the fuel evaporating) before and after, measure the water's temperature rise, and use Q = mcΔT with the mass of the **water**.

Worked example: combustion of ethanol

1. A spirit burner containing ethanol, C_2H_5OH, loses 0.920 g while heating 200 g of water from 19.5 °C to 37.0 °C. Calculate $\Delta_c H$ for ethanol.

2 Energy absorbed by the water (ΔT = 17.5 K):

$$Q = 200 \times 4.18 \times 17.5 = 14\,630 \text{ J}$$

m is the mass of water in the can, not the mass of fuel burned.

3 Moles of ethanol burned, using M_r = 2(12.0) + 6(1.0) + 16.0 = 46.0:

$$n = \frac{0.920}{46.0} = 0.0200 \text{ mol}$$

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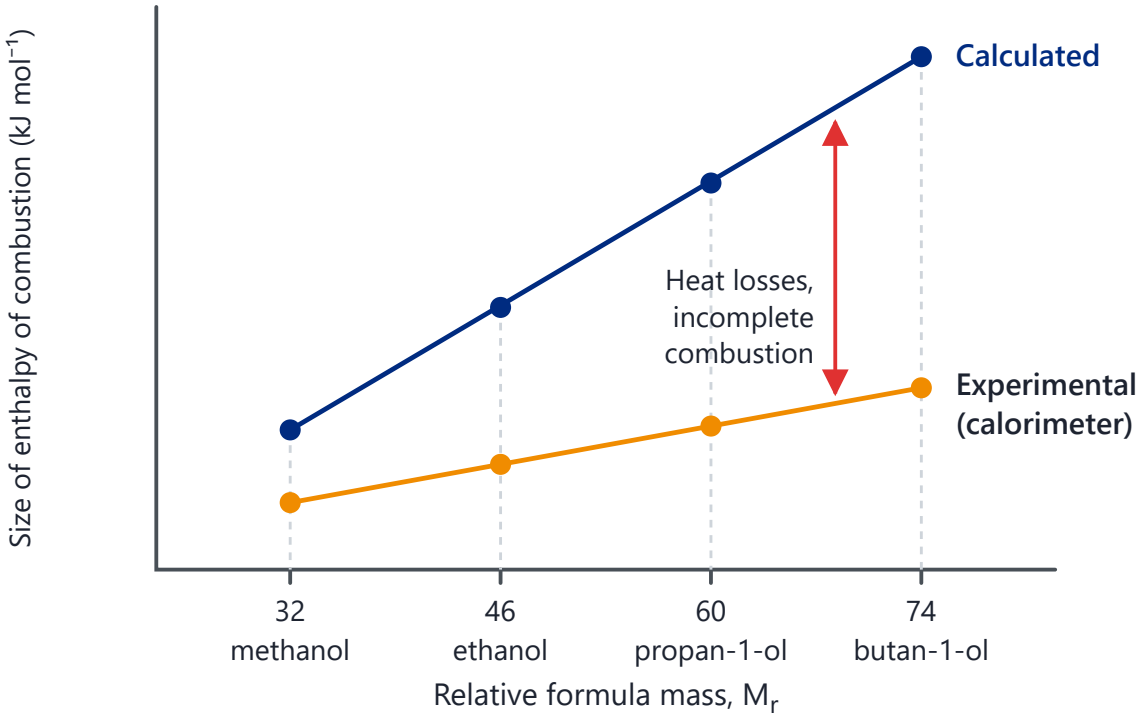
$$\Delta_c H = \frac{-14\,630}{0.0200} = -731\,500 \text{ J mol}^{-1} = -732 \text{ kJ mol}^{-1}$$

The data book value is −1367 kJ mol^{-1}, so only about half of the energy released reached the water.

Combustion calorimetry is much less accurate than the polystyrene-cup method. Most of the error comes from:

- **Heat loss:** hot gases from the flame escape around the can and warm the air instead of the water. A draught shield, a lid on the can and a short gap between flame and can all help.
- **Incomplete combustion:** soot on the bottom of the can shows that some carbon was not burned to CO_2, so less energy was released.
- **Evaporation** of the fuel from the wick between weighings, making the mass "burned" seem too large.
- **The heat capacity of the can** is ignored.
- **Non-standard conditions:** the water formed escapes as steam, not as liquid, so the energy it would release on condensing is lost.

Going up a homologous series, each extra CH_2 group means breaking one more C–C bond, two more C–H bonds and 1½ more O=O bonds, and making two more C=O bonds and two more O–H bonds. The extra amount of energy is the same each time, so $\Delta_c H$ becomes more negative by a roughly constant amount, about 650 kJ mol^{-1} per CH_2: methanol −726, ethanol −1367, propan-1-ol −2021, butan-1-ol −2676 kJ mol^{-1}.



Sketch graph (not to scale) of the size of the enthalpy of combustion against M_r for the alcohols. Both sets of values rise steadily with each CH_2 group, but the calorimeter results are far smaller because of heat losses and incomplete combustion.

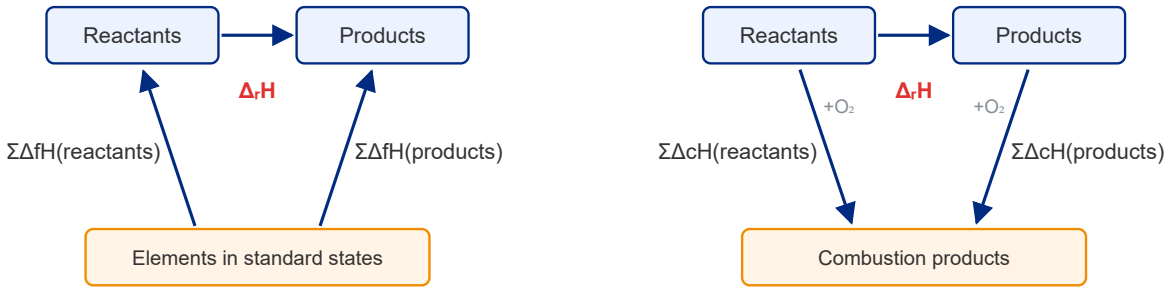
Hess's law

KEY IDEA

Hess's law: the total enthalpy change for a reaction is independent of the route taken, as long as the starting and finishing conditions are the same.

Hess's law follows from the conservation of energy. Imagine two routes from A to B with different enthalpy changes. You could go from A to B by the more exothermic route and come back by the reverse of the other, ending where you started with energy left over. That would create energy from nothing, which never happens.

The law is useful because many enthalpy changes can't be measured directly. You can't make carbon and hydrogen react to give methane in a calorimeter, and you can't heat calcium carbonate while measuring a temperature change. Instead, you build an **enthalpy cycle**: the direct route from reactants to products, plus an indirect route through some other substances whose enthalpy changes are known. The rule for going round a cycle is simple: **follow an arrow and you use its ΔH; go against an arrow and you change its sign**.



Left: a cycle using enthalpies of formation, with arrows pointing up out of the elements. Right: a cycle using enthalpies of combustion, with arrows pointing down into the combustion products (CO_2 and H_2O).

Using enthalpies of formation

Every compound can be made from its elements, so the elements make a convenient bottom corner for the cycle. Going from reactants to products the indirect way means first going backwards down the left arrow (breaking the reactants back into elements, so the sign changes) and then up the right arrow (forming the products). That gives:

$$\Delta_r H = \sum \Delta_f H(\text{products}) - \sum \Delta_f H(\text{reactants})$$

Worked example: reducing iron(III) oxide

1. Calculate $\Delta_r H$ for $Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(s) + 3CO_2(g)$. Data, $\Delta_f H$ / kJ mol^{-1}: $Fe_2O_3(s)$ −824.2, $CO(g)$ −110.5, $CO_2(g)$ −393.5.

2. Sum for the products. Iron is an element in its standard state, so its value is zero. Multiply each value by the number of moles in the equation:

$$\sum \Delta_f H(\text{products}) = 2(0) + 3(-393.5) = -1180.5 \text{ kJ mol}^{-1}$$

3. Sum for the reactants:

$$\sum \Delta_f H(\text{reactants}) = (-824.2) + 3(-110.5) = -1155.7 \text{ kJ mol}^{-1}$$

4. Products minus reactants:

$$\Delta_r H = -1180.5 - (-1155.7) = -24.8 \text{ kJ mol}^{-1}$$

Watch the double negative: subtracting −1155.7 is the same as adding 1155.7.

Using enthalpies of combustion

Enthalpies of combustion are easy to measure for anything that burns, and every organic compound burns to the same products, CO_2 and H_2O. So the combustion products make the bottom corner. This time the indirect route goes down the left arrow (burning the reactants) and then backwards up the right arrow (un-burning the products):

$$\Delta_r H = \sum \Delta_c H(\text{reactants}) - \sum \Delta_c H(\text{products})$$

COMMON MISTAKE

The two formulas have the subtraction the opposite way round. Don't learn them as a rhyme: draw the cycle and follow the arrows, and you can't get the order wrong.

Worked example: the enthalpy of formation of propane

1. Calculate $\Delta_f H$ of propane, $3C(s) + 4H_2(g) \rightarrow C_3H_8(g)$, from $\Delta_c H$ / kJ mol^{-1}: C(s) −393.5, $H_2(g)$ −285.8, $C_3H_8(g)$ −2219.2.

   Carbon and hydrogen won't combine to give just propane, so this can't be measured directly. All three substances burn easily, though.

2. Burn the reactants (down the left arrow):

$$\sum \Delta_c H(\text{reactants}) = 3(-393.5) + 4(-285.8) = -2323.7 \text{ kJ mol}^{-1}$$

3. The product: one mole of propane, $\sum \Delta_c H(\text{products}) = -2219.2 \text{ kJ mol}^{-1}$.

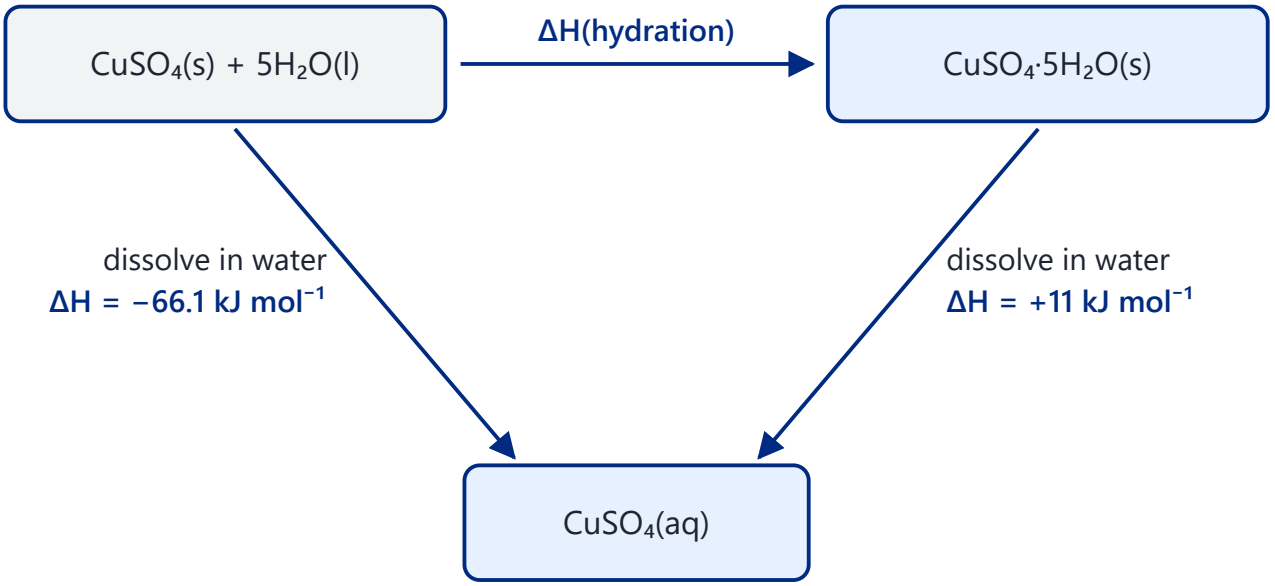
4. Reactants minus products:

$$\Delta_f H(C_3H_8) = -2323.7 - (-2219.2) = -104.5 \text{ kJ mol}^{-1}$$

Finding enthalpy changes that can't be measured directly

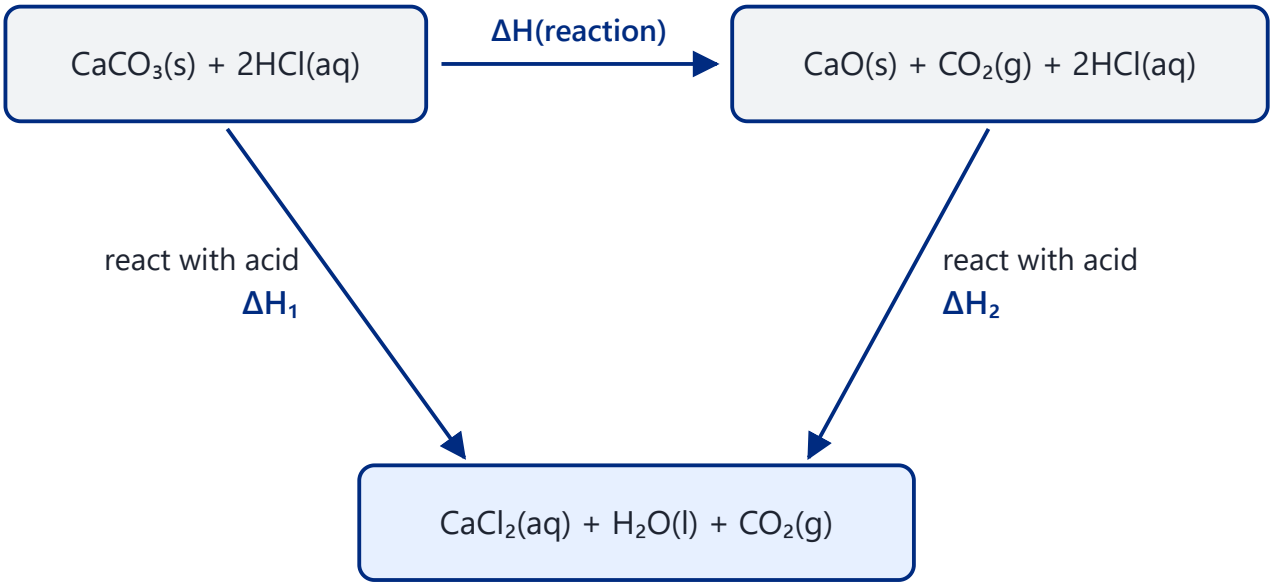
Some reactions are impossible to carry out cleanly in a calorimeter, but you can reach the same products by two reactions that *can* be measured.

Hydrating copper(II) sulfate. White anhydrous copper(II) sulfate turns into blue hydrated crystals when it takes up water: $CuSO_4(s) + 5H_2O(l) \rightarrow CuSO_4{\cdot}5H_2O(s)$. You can't add exactly five moles of water and measure the temperature of a solid. Instead, dissolve each solid separately in plenty of water. Both give the same solution, $CuSO_4(aq)$, so they share a corner of the cycle.



Both salts dissolve to give $CuSO_4(aq)$. Following the arrows, ΔH(hydration) + 11 = −66.1, so ΔH(hydration) = −77.1 kJ mol^{-1}.

Decomposing calcium carbonate. $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ needs strong heating, so you can't measure its heat change with a thermometer. Both calcium carbonate and calcium oxide react with hydrochloric acid to give the same products, and those two reactions can be done in a polystyrene cup. ΔH_1 and ΔH_2 are measured, and $\Delta H = \Delta H_1 - \Delta H_2$. (The CO_2 from the decomposition simply ends up in the bottom box alongside the CO_2 from the acid reaction.)



Calcium carbonate and calcium oxide both react with hydrochloric acid to give the same products. Going round the cycle, $\Delta H(\text{reaction}) + \Delta H_2 = \Delta H_1$.

CORE PRACTICAL

Core Practical 2: determine the enthalpy change of a reaction using Hess's law

A typical version finds the enthalpy change for hydrating anhydrous copper(II) sulfate.

- Pipette 50.0 cm^3 of distilled water into a polystyrene cup standing in a beaker. Record its temperature every minute for three minutes.
- Weigh about 4 g of anhydrous copper(II) sulfate accurately in a dry weighing bottle. At the fourth minute tip it in, stir, and record the temperature every half-minute until about the fifteenth minute. Reweigh the empty bottle to get the exact mass added.
- Repeat with about 6.2 g of hydrated copper(II) sulfate, $CuSO_4 \cdot 5H_2O$, in a fresh 50.0 cm^3 of water. This time the temperature falls.
- Plot both temperature–time graphs, extrapolate to the time of mixing to find ΔT, and work out ΔH for each dissolving per mole of salt.
- **Safety:** copper(II) sulfate is harmful if swallowed and irritates the eyes, so wear eye protection.

Another version uses potassium hydrogencarbonate and potassium carbonate, each reacted with excess hydrochloric acid, to find the enthalpy change of $2KHCO_3(s) \rightarrow K_2CO_3(s) + CO_2(g) + H_2O(l)$. You can run both of those reactions, and both copper sulfate experiments, in the calorimetry simulation above.

Worked example: Core Practical 2 results

1. Experiment 1: 3.99 g of anhydrous $CuSO_4$ (M_r = 159.6) dissolved in 50.0 g of water. Extrapolated ΔT = +7.6 K.

$$Q = 50.0 \times 4.18 \times 7.6 = 1588.4\ \text{J}, \qquad n = \frac{3.99}{159.6} = 0.0250\ \text{mol}$$

$$\Delta H_1 = \frac{-1588.4}{0.0250} = -63.5\ \text{kJ mol}^{-1}$$

2. Experiment 2: 6.24 g of $CuSO_4 \cdot 5H_2O$ (M_r = 249.6) dissolved in 50.0 g of water. Extrapolated ΔT = −1.3 K, so heat was taken in from the water.

$$Q = 50.0 \times 4.18 \times 1.3 = 271.7\ \text{J}, \qquad n = \frac{6.24}{249.6} = 0.0250\ \text{mol}$$

$$\Delta H_2 = \frac{+271.7}{0.0250} = +10.9\ \text{kJ mol}^{-1}$$

A temperature fall means an endothermic process, so this ΔH is positive.

3. Go round the cycle. Route 1: anhydrous salt + water → solution, ΔH_1. Route 2: anhydrous salt + water → hydrated salt (ΔH, unknown), then hydrated salt → solution (ΔH_2).

$$\Delta H + \Delta H_2 = \Delta H_1 \quad \Rightarrow \quad \Delta H = -63.5 - (+10.9) = -74.4\ \text{kJ mol}^{-1}$$

The data book value is about −78 kJ mol^{-1}. The biggest error is in ΔH_2: a fall of only 1.3 K has a large percentage uncertainty.

Bond enthalpies

DEFINITION

Bond enthalpy

The enthalpy change needed to break one mole of a particular covalent bond, with the molecules and the atoms formed all in the **gaseous** state. For example, $H_2(g) \rightarrow 2H(g)$, ΔH = +436 kJ mol^{-1}.

DEFINITION

Mean bond enthalpy

The enthalpy change needed to break one mole of a given type of bond, **averaged over many different**

compounds, with all species gaseous.

Bond enthalpies are always **positive**, because breaking a bond always needs energy. The strength of a bond depends a little on its surroundings. Breaking the four C–H bonds in methane one after another needs four different amounts of energy, because each time the remaining fragment is different. The C–H bond in ethanol is slightly different again. Data books therefore list a **mean** value for each type of bond, such as 413 kJ mol^{-1} for C–H, which is close to, but not exactly, the value in any one molecule.

Bond	Mean bond enthalpy / kJ mol^{-1}	Bond	Mean bond enthalpy / kJ mol^{-1}
H–H	436	O=O	498
C–H	413	O–H	464
C–C	347	C=O (in CO_2)	805
C=C	612	C–O	358
N≡N	945	N–H	391
Cl–Cl	243	H–Cl	432
Br–Br	193	H–Br	366
I–I	151	H–I	298

To estimate ΔH for a reaction between gases, imagine it happening in two stages: break every bond in the reactants to make separate gaseous atoms (energy in), then join the atoms up into the products (energy out). By Hess's law that is the same as the real reaction:

$$\Delta H = \sum(\text{bond enthalpies of bonds broken}) - \sum(\text{bond enthalpies of bonds made})$$

Worked example: burning methane, using mean bond enthalpies

1. Estimate ΔH for $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$. Draw out the molecules to count the bonds: methane has four C–H bonds, each O_2 has one O=O, CO_2 is O=C=O, and each water has two O–H bonds.

2. Bonds broken (endothermic):

$$4(\text{C–H}) + 2(\text{O=O}) = 4(413) + 2(498) = 1652 + 996 = +2648 \text{ kJ mol}^{-1}$$

3. Bonds made (exothermic):

$$2(\text{C=O}) + 4(\text{O–H}) = 2(805) + 4(464) = 1610 + 1856 = 3466 \text{ kJ mol}^{-1}$$

4. $$\Delta H = 2648 - 3466 = -818 \text{ kJ mol}^{-1}$$

Using enthalpies of formation for the same reaction (with water as a gas) gives −802 kJ mol^{-1}. The difference comes from using mean values instead of the actual bond enthalpies in these molecules.

Values from mean bond enthalpies are **less accurate** than values from enthalpies of formation or combustion, for two reasons. The mean values are averages, not the true values for the bonds in these particular molecules. And the method only works for gases: if a substance is a liquid or solid in its standard state, extra energy is involved in separating its molecules, which bond enthalpies leave out. That is why the methane example uses $H_2O(g)$; the standard $\Delta_c H$, which makes $H_2O(l)$, is −890 kJ mol^{-1}.

Working out a bond enthalpy from other data

You can turn the method round and use a known ΔH to find a bond enthalpy. If you need a substance as gaseous atoms and it is not a simple diatomic gas, use its enthalpy of atomisation instead of a bond enthalpy: carbon, C(s) → C(g), needs +717 kJ mol^{-1}.

Worked example: the N–H bond enthalpy in ammonia

1. $\Delta_f H(NH_3(g))$ = −46.1 kJ mol^{-1}. Use E(N≡N) = 945 and E(H–H) = 436 kJ mol^{-1} to find the N–H bond enthalpy. The formation equation is $\frac{1}{2}N_2(g) + 1\frac{1}{2}H_2(g) \rightarrow NH_3(g)$.

2. Bonds broken: half a mole of N≡N and one and a half moles of H–H:

$$\tfrac{1}{2}(945) + 1\tfrac{1}{2}(436) = 472.5 + 654 = 1126.5 \text{ kJ mol}^{-1}$$

3. Bonds made: three moles of N–H. Write the equation:

$$-46.1 = 1126.5 - 3E(\text{N–H})$$

4. Rearrange:

$$3E(\text{N–H}) = 1126.5 + 46.1 = 1172.6 \quad \Rightarrow \quad E(\text{N–H}) = 391 \text{ kJ mol}^{-1}$$

Bond enthalpies also give a feel for how reactive a substance will be. A reaction usually has to start by breaking a bond, so weak bonds (such as the I–I bond, 151 kJ mol^{-1}) break more easily and lead to faster reactions at room temperature than very strong bonds like N≡N (945 kJ mol^{-1}). That is one reason nitrogen is so unreactive.

Quick checks

QUICK CHECK

Which equation represents the standard enthalpy change of formation of ethanol?

- $C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$
- $2C(s) + 3H_2(g) + \frac{1}{2}O_2(g) \rightarrow C_2H_5OH(l)$
- $4C(s) + 6H_2(g) + O_2(g) \rightarrow 2C_2H_5OH(l)$
- $2C(g) + 6H(g) + O(g) \rightarrow C_2H_5OH(l)$

Formation means one mole of the compound made from its elements in their standard states. The third option makes two moles, and the last one starts from gaseous atoms rather than the elements' standard states (C(s), $H_2(g)$, $O_2(g)$). The first is the combustion equation.

QUICK CHECK

25.0 cm^3 of 2.00 mol dm^{-3} HCl is mixed with 25.0 cm^3 of 2.00 mol dm^{-3} NaOH. The temperature rises by 13.0 °C. What is the enthalpy change of neutralisation?

- -27.2 kJ mol^{-1}
- -109 kJ mol^{-1}
- $+54.3$ kJ mol^{-1}
- -54.3 kJ mol^{-1}

The mass is the total of both solutions, 50.0 g, so Q = 50.0 × 4.18 × 13.0 = 2717 J. Moles of water = 2.00 × 0.0250 = 0.0500 mol, so ΔH = −2717 ÷ 0.0500 = −54 340 J mol^{-1} = −54.3 kJ mol^{-1}. Using only 25.0 g gives −27.2; it is negative because the temperature rose.

QUICK CHECK

$\Delta_c H$ of ethene is −1411 kJ mol^{-1} and $\Delta_c H$ of ethanol is −1367 kJ mol^{-1}. What is ΔH for $C_2H_4(g) + H_2O(l) \rightarrow C_2H_5OH(l)$?

- −44 kJ mol^{-1}
- +44 kJ mol^{-1}
- −2778 kJ mol^{-1}
- −330 kJ mol^{-1}

ΔH = $\Sigma\Delta_c H$(reactants) − $\Sigma\Delta_c H$(products) = (−1411 + 0) − (−1367) = −44 kJ mol^{-1}. Liquid water is already fully burned, so its enthalpy of combustion is zero. Subtracting the wrong way round gives +44.

QUICK CHECK

Using mean bond enthalpies H–H = 436, Cl–Cl = 243 and H–Cl = 432 kJ mol^{-1}, what is ΔH for $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$?

- +247 kJ mol^{-1}
- +185 kJ mol^{-1}
- −185 kJ mol^{-1}
- −1111 kJ mol^{-1}

Bonds broken = 436 + 243 = 679; bonds made = 2 × 432 = 864. ΔH = 679 − 864 = −185 kJ mol^{-1}. Forgetting that two moles of H–Cl form gives +247.

QUICK CHECK

Why does a calculation with mean bond enthalpies usually give a slightly different ΔH from a Hess cycle using enthalpies of formation?

- Bond breaking is exothermic, so the signs are reversed.
- Heat is lost to the surroundings during the calculation.
- Hess's law does not apply to gases.
- Mean values are averages over many compounds, not the actual values in these molecules.

Each mean bond enthalpy is averaged over many different molecules, so it is only approximately right for any particular molecule. (Bond breaking is endothermic, and heat loss affects experiments, not calculations.)

QUICK CHECK

Which of these enthalpy changes is always positive?

- Standard enthalpy change of combustion
- Standard enthalpy change of atomisation
- Standard enthalpy change of neutralisation
- Standard enthalpy change of formation

Making gaseous atoms from an element always means breaking bonds, which needs energy. Combustion and neutralisation are always exothermic. Formation is usually exothermic, but some compounds, such as ethene (+52 kJ mol^{-1}), have positive values.

QUICK CHECK

A student measures $\Delta_c H$ of propan-1-ol with a spirit burner and a copper can and gets −1150 kJ mol^{-1}. The data book value is −2021 kJ mol^{-1}. Which is the main reason for the difference?

- The specific heat capacity of water is too large.
- The water in the can was not stirred.
- Much of the heat from the flame escaped to the air instead of heating the water.
- The propan-1-ol contained too much oxygen.

Heat loss from the flame is by far the largest error in combustion calorimetry, with incomplete combustion next. Both make the measured value much less exothermic than the true value.